

IMO(31st) – 1990

First Day

1. Given a circle with two chords AB, CD that meet at E, let M be a point of chord AB other than E. Draw the circle through D, E and M. The tangent line to the circle DEM at E meets the lines BC, AC at F, G respectively. Given that $\frac{AM}{AB} = \lambda$, find $\frac{GE}{EF}$.
2. On a circle, $2n - 1$ ($n \geq 3$) different points are given. Find the minimal natural number N with the property that whenever N of the given points are colored black, there exist two black points such that the interior of one of the corresponding arcs contains exactly n of the $2n - 1$ points.
3. Find all positive integers n having the property that $\frac{2^n + 1}{n^2}$ is an integer.

Second Day

4. Let Q^+ be the set of positive rational numbers. Construct a function $f : Q^+ \rightarrow Q^+$ such that $f(xf(y)) = \frac{f(x)}{y}$ for all $x, y \in Q^+$.
5. Two players A and B play a game in which they choose numbers alternately according to the following rule: At the beginning, an initial natural number $n_0 > 1$ is given, knowing n_{2k} , player A may choose any $n_{2k+1} \in N$ such that $n_{2k} \leq n_{2k+1} \leq (n_{2k})^2$. Then player B chooses a number $n_{2k+2} \in N$ such that $\frac{n_{2k+1}}{n_{2k+2}} = p^r$, where p is a prime number and $r \in N$. It is stipulated that player A wins the game if he (she) succeeds in choosing the number 1990, and player B wins if he (she) succeeds in choosing 1. For which natural number n_0 can player A manage to win the game, for which n_0 can player B manage to win, and for which n_0 can players A and B each force a tie?
6. Is there a 1990-gon with the following properties (a) and (b)?
 - (a) All angles are equal.
 - (b) The lengths of the 1990 sides form a permutation of the numbers $1^2, 2^2, \dots, 1989^2, 1990^2$.

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